

Lecture 5, Jan 30-2026

Recall that:

$$\frac{d}{dx}(\log_b(x)) = \frac{1}{x \ln b}$$

Example. Differentiate $f(x) = \log_{10}(2 + \sin x)$.

Sln.

$$f'(x) = \frac{1}{(2 + \sin x) \ln(10)} \cdot \frac{d}{dx}(2 + \sin x) = \frac{\cos x}{\ln(10)(2 + \sin x)}$$

$$\frac{d}{dx}(b^x) = b^x \ln(b)$$

Example. $\frac{d}{dx}(10^{x^2})$.

Sln.

$$\frac{d}{dx}(10^{x^2}) = 10^{x^2} \ln(10) \cdot \frac{d}{dx}(x^2) = 2 \ln(10) x 10^{x^2}$$

$$\int b^x dx = \frac{b^x}{\ln(b)} + C, \quad b \neq 1.$$

Example.

$$\int_0^5 2^x dx = \left. \frac{2^x}{\ln(2)} \right|_0^5 = \frac{2^5}{\ln(2)} - \frac{2^0}{\ln(2)} = \frac{31}{\ln(2)}$$

Problem. Compute y' if $y = \frac{x^{3/4} \sqrt{x^2 + 1}}{(3x + 2)^5}$.

Steps in Logarithmic Differentiation.

1. Take natural logarithms of both sides of an equation $y = f(x)$, and use the laws of logarithms to expand the expression.
2. Differentiate implicitly with respect to x .
3. Solve the resulting equation for y' and replace y by $f(x)$.

Sln. 1.
$$\begin{aligned}\ln(y) &= \ln\left(\frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}\right) \\ &= \ln(x^{3/4}\sqrt{x^2+1}) - \ln((3x+2)^5) \\ &= \ln(x^{3/4}(x^2+1)^{1/2}) - 5\ln(3x+2) \\ &= \ln(x^{3/4}) + \ln((x^2+1)^{1/2}) - 5\ln(3x+2) \\ &= \frac{3}{4}\ln(x) + \frac{1}{2}\ln(x^2+1) - 5\ln(3x+2).\end{aligned}$$

2.
$$\begin{aligned}\frac{d}{dx}(\ln(y)) &= \frac{d}{dx}\left(\frac{3}{4}\ln(x) + \frac{1}{2}\ln(x^2+1) - 5\ln(3x+2)\right) \\ \frac{y'}{y} &= \frac{3}{4}\frac{d}{dx}(\ln(x)) + \frac{1}{2}\frac{d}{dx}(\ln(x^2+1)) - 5\frac{d}{dx}(\ln(3x+2)) \\ &= \frac{3}{4x} + \frac{1}{2} \cdot \frac{2x}{x^2+1} - 5 \cdot \frac{(3)}{3x+2} \\ &= \frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}.\end{aligned}$$

3. Solving for y' .

$$y' = \left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right) \boxed{y}$$

$$y' = \left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right) \cdot \boxed{\frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}}.$$

Example. Differentiate $y = x^{\sqrt{x}}$

Sln.

$$\ln(y) = \ln(x^{\sqrt{x}}) = \sqrt{x} \ln(x)$$

$$\frac{d}{dx}(\ln(y)) = \frac{d}{dx}(\sqrt{x} \ln(x))$$

$$\frac{y'}{y} = \frac{1}{2\sqrt{x}} \ln(x) + \frac{\sqrt{x}}{x} = \frac{\ln(x)}{2\sqrt{x}} + \frac{1}{\sqrt{x}} = \frac{\ln x + 2}{2\sqrt{x}}$$

$$y' = \left(\frac{\ln(x) + 2}{2\sqrt{x}} \right) \cdot x^{\sqrt{x}}$$

The number e as a limit.

We know that $f(x) = \ln(x)$ $f'(x) = 1/x$ then $f'(1) = 1$. We now use this fact to express the e number as a limit.

$$\begin{aligned} 1 = f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{x \rightarrow 0} \frac{f(1+x) - f(1)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\ln(1+x) - \ln(1)}{x} \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x) \\ &= \lim_{x \rightarrow 0} \ln \left((1+x)^{1/x} \right) \end{aligned}$$

So,

$$\lim_{x \rightarrow 0} \ln \left((1+x)^{1/x} \right) = 1$$

$$\begin{aligned} e = e^1 &= e^{\lim_{x \rightarrow 0} \ln \left((1+x)^{1/x} \right)} \\ &= \lim_{x \rightarrow 0} e^{\cancel{\ln} \left((1+x)^{1/x} \right)} \\ &= \lim_{x \rightarrow 0} (1+x)^{1/x} \end{aligned}$$

$$e = \lim_{x \rightarrow 0} (1+x)^{1/x}$$

let $n = \frac{1}{x}$. when $x \rightarrow 0^+$ $n \rightarrow \infty$ then.

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

6.5. Exponential Growth and Decay.

Goal. let $y(t)$ is the value of a quantity y at time t and if the rate of change of y with respect to t is proportional to its size $y(t)$ at any time, then

$$\frac{dy}{dt} = \kappa y \quad (I)$$

where κ is a constant. The equation (I) sometimes called the law of natural growth ($\kappa > 0$) or the law of natural decay ($\kappa < 0$).

Thm. The only solution of the differential equation $dy/dt = \kappa y$ are the exponential functions

$$y(t) = y(0)e^{\kappa t}.$$